

Day 7 Cheatsheet

Data Visualization

Major concepts:

- The **ggplot2** package is a tidyverse package for producing graphics.
- Tidy data: Each variable forms a column and each observation forms a row.
- Having data in tidy format and long format makes creating plots easier!
- Remember to use a + at the end of each line when adding new layers to plots.

Pipes like `|>` or `%>%` do not work to add layers to plots, but can be used to pipe data into plots.

This works: `ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_boxplot()`

This works: `iris |> ggplot(aes(x = Species, y = Petal.Length)) + geom_boxplot()`

This doesn't: `iris |> ggplot(aes(x = Species, y = Petal.Length)) |> geom_plot()`

- Factor: A factor is a special character vector where the elements have pre-defined groups or “levels”. You can think of these as qualitative or categorical variables. An example is 1st grade, 2nd grade, etc. It is important to specify factors as the class **factor** so that R recognizes it as such.

Functions

Library/Package	Piece of code	Example of usage	What it does
ggplot2	<code>ggplot()</code>	<code>ggplot(data = iris)</code>	Begins a plot that is finished by adding layers.
ggplot2	<code>aes()</code>	<code>ggplot(data = iris, aes(x = Species, y = Petal.Length))</code>	Aesthetic Mappings - Designates how variables in the data object will be mapped to the visual properties of the ggplot.
ggplot2	<code>geom_boxplot()</code>	<code>ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_boxplot()</code>	Creates a boxplot when added as a layer to a <code>ggplot()</code> object.
ggplot2	<code>geom_density()</code>	<code>ggplot(data = iris, aes(x = Petal.Length)) + geom_density()</code>	Creates a smoothed density plot when added as a layer to a <code>ggplot()</code> object based on the computed density estimate.
ggplot2	<code>geom_point()</code>	<code>ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_point()</code>	Creates a scatterplot when added as a layer to a <code>ggplot()</code> object.

Library/Package	Piece of code	Example of usage	What it does
ggplot2	geom_line()	ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_line()	Creates a line plot when added as a layer to a <code>ggplot()</code> object by connecting the points in order of the x axis variable.
ggplot2	geom_hline()	ggplot(data = iris, aes(x = Species, y = Petal.Length)) + geom_point() + geom_hline(yintercept = 4)	Annotates a plot with horizontal lines when added as a layer to a <code>ggplot()</code> object with one of the <code>geom</code> functions used to draw the plot, for example, <code>geom_point()</code> .
ggplot2	theme_classic()	ggplot(data = iris, aes(x = Petal.Length)) + geom_density() + theme_classic()	Displays <code>ggplot</code> without grid lines. The <code>ggtheme</code> documentation has descriptions on additional themes that can be used.
ggplot2	xlab(); ylab(); ggtitle()	ggplot(data = iris, aes(x = Petal.Length)) + geom_density() + xlab("Petal Length")	Modifies the labels on the x axis and on the y axis, and sets the title of a <code>ggplot</code> , respectively.
ggplot2	facet_grid()	ggplot(data = iris, aes(x = Petal.Length)) + geom_density() + facet_grid(~Species)	Plots individual graphs using specified variables to subset the data and create plots in a grid shape.
ggplot2	facet_wrap()	ggplot(data = iris, aes(x = Petal.Length)) + geom_density() + facet_wrap(~Species, scales = "free", nrow = 2)	Plots individual graphs using specified variables to subset the data. Allows for different shapes and for different axes scales.
ggplot2	ggsave()	ggsave(filename = "plotname.pdf")	Saves the last plot in working directory.

[Click here for a summary of the ggplot2 theme system~](#)

Factors

Library/Package	Piece of code	Example of usage	What it does
base	<code>factor()</code>	<code>fact_qual <- factor(c("poor", "fine", "good"))</code>	Creates a factor out of a vector. Levels are ordered by default alphanumerically.
base	<code>levels()</code>	<code>levels(fact_qual)</code>	Shows the levels (and order) of a factor vector.
forcats	<code>as_factor()</code>	<code>fact_qual <- as_factor(c("poor", "fine", "good"))</code>	Creates a factor out of a vector. Levels are ordered by default in order of first appearance in the data.
forcats	<code>fct_reorder()</code>	<code>ggplot(iris, aes(x = fct_reorder(Species, Sepal.Width, mean), y = Sepal.Width)) + geom_boxplot()</code>	Allows the order of the levels of a factor variable to be changed according to another variable. In this case Species is ordered by the mean of Sepal.Width. This is especially helpful for plots or statistical output!

* This format was adapted from the cheatsheet format from AlexsLemonade.